Water management in Barcelona

A XXI century solution for drought resilience

EE4C11 Systems Engineering

Resilient Cities Group 7

Delft University of Technology

TU Delft

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A XXI century solution for drought resilience

by

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Cover: Sau Reservoir, in the metropolitan region of Barcelona, at 18% of its capacity: a few visitors at the church of the former village of Sant Romà de Sau, normally submerged under the water. By Albert Salamé, November 2022.

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Preface

Al meu país la pluja no sap ploure
O plou poc o plou massa
Si plou poc és la sequera
Si plou massa és la catàstrofe

In my country, rain doesn't know how to rain
It either rains too little or too much
If it rains too little, it's a drought
If it rains too much, it's a catastrophe

Excerpt from the lyrics of the famous song "Al meu país la pluja",
by the Spanish singer-songwriter Raimon.

Resilient Cities Group 7
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1. Problem Statement

1.1. Introduction

A resilient city is defined by the European Commission as one that can "maintain a continuity of services and functions throughout any shock or stress, while protecting and enhancing people's lives" [1]. Resilient cities are becoming increasingly relevant, and their development is bounded by two global factors:

- Projections from the World Bank [2] suggest that the world's urban population will reach 70% by 2050, effectively doubling the size of cities compared to the present. This change is driven by the promise of an improved quality of life and numerous opportunities in large urban centers.
- Climate change threatens the environmental resources that sustain societies and directly endangers the well-being of citizens through increasingly severe environmental disasters.

This project addresses the intersection of these two critical factors: a solution for the challenge posed when resource constraints, driven by climate change, intersect with a growing population accustomed to abundance.

1.2. Location and motivation

Due to the challenges ahead, but also the opportunities these can represent, Barcelona has great potential to become a Resilient City. The city of Barcelona faces several environmental challenges, including flooding during heavy rains, heatwaves, prolonged droughts, wildfires, and coastal erosion. Climate change exacerbates these issues, with predictions of more severe and frequent extreme weather events [3]. Water management plays a pivotal role in addressing these challenges.

Barcelona has already taken proactive steps in flood management [4] and wildfire risk prevention. On the other hand, in recent months, the city has come out of its longest recorded drought, lasting 32 months, which together with an outdated infrastructure [5], has strained water resources to unprecedented levels. Restrictions on water use have been enforced, impacting public services and highlighting the need for innovative solutions. Traditional approaches like desalination, river diversions, and reservoir construction are costly and environmentally risky.

1.3. Current state and need

To solve this problem, it is important to delve into the logistics of the city's current water infrastructure. The city consumes an average of 106 liters per person per day [5], one of the lowest in Europe, totaling around 130 hm^3 annually in the metropolitan area (63 hm^3 for the single municipality). With the City Council aiming for consumption below 100 liters per person, there is still room for improvement. This is why the Catalan Water Agency has put in motion a 2.4 billion plan to adapt to new periods of scarcity [5].

Drinking water sources include the Ter and Llobregat river basins, with reservoirs with maximum capacity of 700 hm^3 [6]. Two desalination plants further supplement the water supply, with the Llobregat plant being Europe's largest for urban use, contributing 60 hm^3/year [7]. Non-potable water comes from aquifers: the abandonment of industrial wells has raised groundwater levels, allowing for groundwater utilization of 4 hm^3/year [7]. Reclaimed water from sewage treatment plants is another non-potable source, while rainwater capture is being explored for various uses.

In total, 58% of the water resources consumed by Barcelona is generated from desalination or reuse, while 42% comes from rivers and wells [5]. Its dependence on rivers and wells makes the city extremely vulnerable to environmental conditions. On the other hand, 490 hm^3 treatment plant wastewater is yearly discharged into the sea in Catalonia [5]: 201 out of the 548 (37%) wastewater treatment plants are in the Barcelona province, and 113 (20%) in the metropolitan area [8]. This wastewater could well be fed back into the network (as it is being done in Singapore [9]), at a household, building or district level. There are nowadays no other alternatives to unreliable conventional surface water sources, which is why there is a clear need for innovation. In this project, we will outline our comprehensive approach to address Barcelona's water scarcity, emphasizing collaboration, innovation, and community engagement in our pursuit of enhanced water security and urban resilience.

1.4. Problem statement

To address these issues, this project narrows its focus to water management in Barcelona, emphasizing on sustainability and viability. Thus, the problem statement is: **How could we improve the preparedness for water stress events in Barcelona?**

2. System Definition Matrix

A system can help alleviate the water restrictions imposed by reducing the water consumption in several or specific activities. How effective it is in doing so can be measured by the reduction in percentage of water consumption compared to the baseline usage. Such a system can be achieved by working on the infrastructure and distribution network of the water in the city, adopting technological solutions and changing the population consumption behavior. Naturally, this will be limited by the legality, technology and weather conditions among others. But once put in place, the effectiveness of the system will be apparent in the water consumption levels and other variables like volume of water saved through circular practices.

The description of one facet of the solution for improving the water management of a city like Barcelona has been given in the previous paragraph. This is better conveyed via the System Definition Matrix (SDM), a tool that facilitates the decomposition of the problem statement, allowing a more profound understanding of the issue and defining the scoping and bounding of the project. A small subtable of the complete SDM for the problem statement is given in Table 1, transposed for better readability. The complete SDM can be found in Appendix A.

Needs	Fulfill the water requirements and avoid water restrictions	Available for use as soon as possible
Objectives	Reduce water consumption	Based on available technology
Criteria	- Percentage reduction in water consumption compared to the baseline - Actual reduction in water consumption in cubic meters - Percentage increase in water recycling or reuse - Reduction in water wastage through circular consumption practices	- Use of technology that is currently accessible and deployable - Avoidance of reliance on technologies that are not yet mature or readily available
Parameters	- Infrastructure design - Distribution network - Technology adoption - Population consumption behavior - City buildings and infrastructure	- Water treatment technology - Infrastructure design - Technology adoption - Distribution network - Funding sources
Variables	- Water consumption levels - Water-use efficiency ratio - Percentage of recycled or reused water - Volume of water saved through circular practices	- Technology readiness - Market availability - Cost of technology
Constraints	- Legal limitations - Weather conditions - Minimum necessary water per capita - Citizen noncompliance - Current infrastructure - Technological limitations - Population	- Budget - Health and safety regulations - Market constraints - Scalability constraints - Environmental limitations - Citizen noncompliance - Time dedicated to development - Technological limitations

Table 1: System Definition Matrix

3. Concept Generation

In the following sections the process for generating concepts is outlined and the generated concepts to solve the problem stated in section 1 are described and analysed.

Concept Descriptions

A multitude of concepts were generated in a series of intensive brainstorming sessions, using creativity techniques like law-breaker and semantic intuition. These were used to encourage some out of the box ideas and prevent a majority of concepts getting railroaded in a singular direction. Listed below are the most intriguing and promising avenues the team came up with.

3.1. Water reusing/re-cycling system at household level

Barcelona as a city plans on reducing the average household daily water consumption to 100 litres per day by 2030 [5]. This target can only be reached by the introduction and implementation of individual household level water reuse and recycling techniques. The city is heavily interested in emerging technologies in the domain of gray water reuse and recycling among others, and is highly likely to adopt policies and promote solutions in the space.

3.2. Reducing evaporation water loss in current reservoirs

A majority of water storage structures in Barcelona and its vicinity are above ground and open air. Evaporation from standing bodies of water has become a major source of water loss and cause for concern. It is thus essential for a city on the brink of water scarcity to ensure that its current infrastructure is able to retain the little water it gathers.

3.3. Water-collecting green building: green roofs to collect water

Green buildings as a concept refers to the efforts to make pre-existing infrastructure more sustainable as well as create new infrastructure with a strong focus on resource conservation and generation. Among various solutions, the ones to have the largest long-term impacts are the ones focused on replenishment and sustainability. This is where the idea of green roofs truly shines [10–12].

3.4. Expanding and building reservoirs

The focus is on current resource expansion over efficient resource utilization. This can be achieved in a combination of methods including expanding pre-existing reservoirs, creating more desalination plants or even trying to convert the groundwater table of Barcelona into a replenishable and reliable source (via permeable asphalt etc.).

3.5. Sensitivity and awareness campaign

The idea is to inculcate a social movement that is in harmony with technical solutions. Proper sensitization and awareness of the populace to the problem of water scarcity can be an immense help in reducing current water usage rates and behaviors while simultaneously making them more open to adopting new and pre-existing solutions. It is necessary to remember that Barcelona is already one of the European cities with the lowest water consumption, and that after the water crisis of 2008, there has been a certain awareness of conservation among the population.

4. Trade-off table

Trade-off tables are useful tools for the evaluation of different system options and comparing them based on certain criteria. Ten of the most important criteria were chosen for evaluation and divided into three categories, namely Desirability, Feasibility, and Viability.

A 4-point scoring system was utilized for each concept per criteria. This score was multiplied by the given weight of the criteria (0,8/ 1,0/ 1,2). The limited precision of the scores and this weighting system align with the inherent challenge of quantifying such magnitudes, as well as the subjective nature of personal assessments: a broader range of variability add an illusion of complexity, but beyond the quantification noise, it would not provide more precision than the results presented. Regarding the weighting, the ten selected criteria share the same high level of relevance (that's why they were chosen), and the weightings have been added to distinguish between three subjective levels of prioritization, chosen to avoid underestimating or over-prioritizing any criteria, while also adding nuance to the assessments.

Listed below are some sample criteria, chosen to justify the associated weightings and provide an understanding of why each criterion has been assigned the selected level of priority. The complete trade-off table can be found in Appendix A.

Reasonable scale, budget and technical complexity (1,2)

No matter the ingenuity of a solution, its success in the public sphere is dependent on its feasibility and hence the scale, budget, and complexity of a project is an essential criterion of evaluation.

Low environment footprint (1,2)

It is essential that any concept in consideration not create its own set of problems in the future. Any solution that results in a large negative environmental impact defeats the purpose of the implementation in the first place.

Reduction in water consumption (1)

While a general reduction in water consumption could potentially be beneficial, it may not necessarily be a part of the solution to the problem. Ensuring a cost-effective implementation or a low environmental impact takes precedence. On the other hand, this criteria should be prioritized over the residents' inconveniences.

Minimization of inconveniences to residents (0,8)
It would be desirable to have a solution that results in no inconvenience to its residents, but that does not take precedence over the functionality of the solution, i.e. help to reduce and alleviate water stress events in the city.

Category	Criteria	Weight	3.1	3.2	3.3	3.4	3.5
Desirability	Reduction in water consumption	1	4	1	3	1	4
	Long-term water availability	1	3	3	4	3	2
	Improvement of the current systems	1	2	4	1	3	1
	Increase in water recycling or reuse.	1	4	1	3	1	2
Feasibility	Scale, budget and complexity	1,2	2	4	1	2	4
	Reduction in interruption of supply	0,8	3	3	4	3	3
	Minimization of inconveniences	0,8	2	4	3	3	1
Viability	Cost-effectiveness	1	2	4	2	2	3
	Returns on investment	1	3	3	3	2	1
	Low environmental footprint	1,2	4	2	4	2	4
	Overall	10	**29,2**	**28,8**	**27,6**	**21,6**	**25,8**

Table 2: Trade-off table, compressed

5. Requirements

We would like to have a deep dive into the requirements that come with our concepts, for that we delineated project requirements for the three concepts that scored the most in our trade-off table as shown in Table 3. These requirements have been categorized into four domains: 1) system design requirements, 2) functionality requirements, 3) performance requirements, and 4) qualitative requirements. This systematic approach ensures a comprehensive assessment and serves as a foundational framework for the subsequent phases of project development and evaluation.

A	Water reusing/recycling system in household level
B	Reducing evaporation water loss in current reservoirs
C	Water-collecting green building: green roofs to collect water

Table 3: Selected Concepts

In the context of our systems engineering project, we have compiled a comprehensive list of requirements for all three conceptual designs. However, due to space limitations, we will present the detailed requirements for Concept B in this main document, while the requirements for the other two concepts can be found in Appendix A.

6. Conflicts analysis

In the preceding sections of this document, we examined the sometimes conflicting requirements for the selected concepts. Due to constraints in document length, we have chosen to present the conflict analysis of Concept B in the main document, the complete conflict analysis of concepts A and C can be found in Appendix A..

In the conflict analysis of Concept B, we explore the trade-off between sometimes non-compatible needs that a project would imply. The negotiation between the conflicting requirements is crucial for the success of the project. These conflicts demonstrate the complex challenges faced by stakeholders in water reservoir management, offering valuable insights into the research for solution for sustainable and efficient solutions reservoir management

The Concept B has the following conflicting requirements:

1. **RQB9 (Energy Efficiency) vs. RQB6 (Real-time Monitoring):**

Constant monitoring and data transmission can be energy-intensive, leading to increased operational costs. Finding a balance between real-time monitoring and energy efficiency is essential.

Design requirements	
RQB1	The system should not require significant modification on the current reservoir system.
RQB2	The design should consider the local ecosystem and minimize any potential negative impacts on the environment.
RQB3	The system should be scalable to accommodate reservoirs of larger size
RQB4	The design should facilitate easy access for maintenance and repairs to ensure the system's continuous operation.
Functionality requirements	
RQB5	The system must effectively reduce water evaporation in the reservoir by a specified percentage.
RQB6	It should provide real-time data and feedback on water levels, system status, and evaporation rates.
RQB7	The system should allow remote monitoring and control to optimize water conservation efforts.
RQB8	Ensure that the system maintains water quality and prevents the buildup of harmful substances.
RQB9	The system should be designed for energy efficiency to minimize operational costs.
Performance requirements	
RQB10	The system should maintain the evaporation rate below a defined threshold, even under varying weather conditions.
RQB11	The project should result in a quantifiable reduction in water loss due to evaporation.
RQB12	The system should have a rapid response time to changing weather conditions or reservoir usage.
RQB13	The monitoring system should provide highly accurate data on water levels and evaporation rates.
Qualitative requirements	
RQB14	The system should have an intuitive and user-friendly interface for reservoir operators and maintenance personnel.
RQB15	Promote community awareness and engagement regarding water conservation efforts and the project's benefits.
RQB16	Ensure that the project complies with all relevant local, state, and federal regulations and permits.

Table 4: List of requirements for concept B

2. **RQB14 (User-Friendly Interface) vs. RQB7 (Remote Control):**

A highly user-friendly interface might be more complex to design and operate, potentially conflicting with the ease of remote control.

3. **RQB5 (Water Savings) vs. RQB8 (Water Quality Maintenance):**

Reducing water evaporation can involve covering the reservoir, but this may lead to reduced oxygenation and increased algae growth. Finding a way to save water without negatively impacting water quality can be technically complex and costly.

4. **RQB16 (Regulatory Compliance) vs. Cost-effectiveness:**

Adhering to regulations is essential, but it can strain project budgets. Balancing compliance with cost-effectiveness may require innovative solutions and negotiations with regulatory authorities.

The conflict matrix is shown as follows:

	RQB1	RQB2	RQB3	RQB4	RQB5	RQB6	RQB7	RQB8	RQB9	RQB10	RQB11	RQB12	RQB13	RQB14	RQB15	RQB16
RQB1																
RQB2																
RQB3																
RQB4																
RQB5																
RQB6																
RQB7																
RQB8					C											
RQB9						C										
RQB10																
RQB11																
RQB12																
RQB13																
RQB14							C									
RQB15																
RQB16																

Note: RQB16 (Compliance) would pose constraints on the cost-effectiveness of the project

Figure 1: Conflict Matrix of Concept B

In conclusion, the conflicts we have found out underscore the complexities of the decision-making process. It is important to balance these trade-offs and make compromises. Finding an equilibrium between different requirements is not just a challenge but a necessity. Sustainable and effective solutions invariably require thoughtful navigation through these intricate landscapes, where compromise is often the key to achieving harmony between conflicting objectives, and where innovation and negotiation become indispensable tools for progress.

7. Key Requirements

After making an exhaustive list of requirements in the areas of design, functionality, performance, and qualitative characteristics, 5 key requirements have been selected for an efficient and sustainable water management system to manage the drinking water problem in Barcelona.

The design requirements prioritize RQA4, emphasizing the need for consistent water pressure to ensure seamless water distribution—a fundamental aspect of the system's functionality [13]. Shifting to functionality requirements, RQA18 highlights a low maintenance approach, specifying routine tasks like filter replacement or cleaning occurring no more than once every six months, ensuring system longevity and reducing operational costs. Within performance requirements, RQA21 takes precedence, emphasizing the system's adaptability to various plumbing configurations [14] [15] and water sources, bolstering its effectiveness across Barcelona's diverse water infrastructure. Lastly, in qualitative requirements, RQA24 introduces compatibility with smart home automation systems, enabling users to remotely control and monitor the water system, enhancing convenience and user experience.

These identified requirements align with the goals of the stakeholders involved in water management. For sewage management companies, the emphasis on consistent water pressure and low maintenance in the design and functionality requirements resonates with their aim to ensure smooth sewage operations. Private owners benefit from the adaptability and smart technology integration, enhancing control and monitoring capabilities. Building regulators find assurance in the system's versatility and compatibility with diverse plumbing configurations, aligning with regulatory objectives. Furthermore, these requirements contribute directly to the objectives of feasibility, ensuring practical implementation, and reliability, guaranteeing the consistent and dependable performance of the water management system.

Number	Requirement	Relevant stakeholder	Relevant objective
RQA1	The system must be simple to install*	Sewage management company / workers	Feasibility
RQA4	The system shall maintain adequate water pressure	Owner/residents of the building	Reliability
RQA18	The system should have a low maintenance requirement, with routine maintenance tasks such as filter replacement or cleaning needed no more than once every six months**.	Owner/residents of the building	Feasibility
RQA21	The system should be adaptable to various household plumbing configurations and water sources, including municipal water, well water, and rainwater harvesting systems.	Owner/residents of the building	Reliability
RQA24	The system should be compatible with smart home automation systems, allowing users to control and monitor the system remotely through smartphones or other smart devices.	Building regulations makers	Reliability

Table 5: List of key requirements, relevant stakeholders and relevant objectives

* It should not exceed the complexity level of a typical plumber's job. Specialized workers should not be required for the installation.

** This time frame falls within a reasonable range for replacements. Ideally, in terms of environmental concerns, efficiency, and user convenience, this duration could be reduced to once a year. Nevertheless, for a system in developmental stages, it is wiser to provide a safety margin.

8. Functional Decomposition

The functional decomposition table [16] presented here offers a systematic dissection of key functions related to house-hold level system for water management in Barcelona. At its core, six high-level functions are identified,

encompassing tasks such as efficient water distribution, and proper collection and storage of water. Each of these primary functions undergoes a detailed breakdown, branching into lower-level functions. This hierarchical structure provides a complete comprehension of the system's functionalities.

The functional decomposition has been very comprehensive, as can be seen in Appendix A. However, in this document, as well as for a more manageable Morphological Chart, only the sub-functions up to a certain level of division are presented in Table 6.

Higher level parent function	**Sub-high level functions**
Collect and store water [17]	Collect greywater
	Take fresh water from grid
	Store greywater
Maintain water quality [17] [18]	Disinfect water
	Filter water
	Neutralize pH level
Monitor water quality	Monitor the pH levels
	Monitor the turbidity
	Monitor the temperature
	Monitor the total dissolved solids
	Monitor the dissolved oxygen levels
	Monitor chlorine levels
	Monitor algae proliferation
Distribute water	Control global water pressure
	Monitor water pressure
	Supply filtered water
	Flush recycled water/waste into the sewage system
	Leakage monitoring
Offer high compatibility and versatility to the user	Be installable in small utility rooms or spaces
	Be compatible with various household plumbing configurations
	Allow remote control and monitoring
	Be compatible with different power sources
Enable user interaction	Provide user monitoring interface
	Notify maintenance
	Provide automated self-service guidelines
	Auto-verify successful system installation
Ensure user comfort	Operate quietly to avoid disruption of household activities
	Reduce maintenance needs
	Reduce the odor of greywater
	Ensure water supply

Table 6: Functional decomposition table: Higher level functions and sub-high level functions

In the complete functional decomposition (Appendix A), one can also find functions hierarchically organized in tree diagrams, such as the one in Figure 2. Due to space constraints, not all of them could be included to this document.

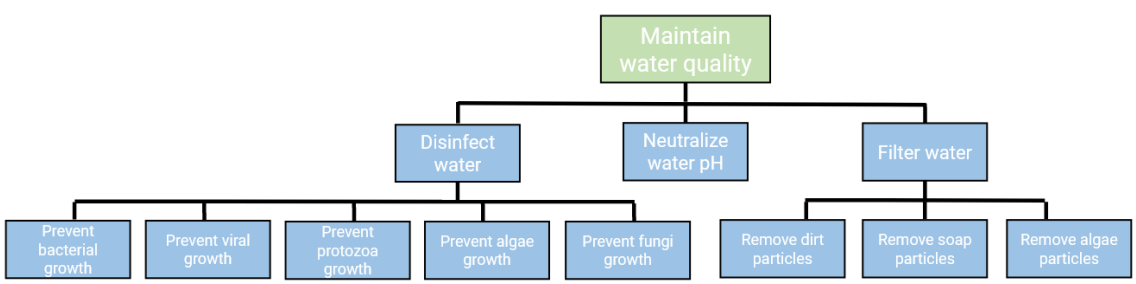

Figure 2: Function decomposition of the high-level function "Maintain water quality".

For example, regarding the "Maintain water quality" function, as shown in Figure 2, which is crucial for the proper

functioning of our system, three sub-functions of the same hierarchy can be derived, namely ”Disinfect water”, ”Neutralize water pH”, and ”Filter water”. The former, related to the elimination of parasites and other health-threatening elements, can be further divided into each of the specific targets the system aims to combat. The same applies to the latter, which aims to filter various components within the water. Note that for very low-level functions, the means will be the same or very similar: this is why only medium level functions are studied in the Morphological Chart.

9. Morphological Chart

After exploring the functional space, a morphological chart [19] was used to systematically explore and define the different possible combinations of functions and means for achieving them. The goal is to create a solution space from which multiple alternate solutions can be generated.

9.1. Generation

It was decided to focus on the 2nd-level functions (second highest level) for generating the means. These functions provide the abstraction of the higher level functions which allows the space for the possibility of multiple solutions while also providing the granularity of lower level functions which allows for the generation of many actionable and effective means. Provided below is a subset of the functions for the morphological table. The complete list can be found in Appendix A.

Higher level parent function	Function for which means are generated
Maintain water quality	Disinfect water
	Filter water
	Neutralize pH level

Table 7: A Subset of functions in focus

For each of the functions, 9-10 means were provided. A concerted effort was made to offer a diverse set of means including pre-existing established technologies as well as emerging solutions. The complete morphological table can be found in Appendix A, shown below is a subset of the table for the functions under the high-level function ”Maintain water quality”.

Function	Means				
Disinfect water	UV disinfection	Chlorination [20]	Chloramination	Ozonation	...
Filter water	Sediment filtration	Reverse Osmosis	Microfiltration	Ultrafiltration	...
Neutralize pH level	pH buffering agents	Calcium hydroxide	Limestone	Alkaline Filter Media	...

Table 8: A subset of the morphological table at the high-level function ”Maintain water quality”.

9.2. Means criteria and eliminated means

Means need to be pruned to provide a workable solution space. 10 criteria were used to compare and score each mean. The list of criteria used and the selected and eliminated means can be checked in the Morphological Chart at Appendix A. Below is a small subset of means eliminated and the justifications for their elimination.

Ozonation

High-level function: Maintain water quality
Function: Disinfect water
Ozonation is not a practical choice for disinfecting water in a personal gray water plumbing system mainly due to its technical complexity [21] and the potential safety risks associated with ozone gas. Implementing an ozone treatment system requires specialized equipment and expertise, making it cost-prohibitive and impractical for personal use.

Reverse Osmosis

High-level function: Maintain water quality
Function: Filter water
Reverse osmosis systems are typically designed for purifying drinking water and are not well-suited for handling the diverse contaminants found in gray water. They are highly inefficient, wasting a significant amount of water during the filtration process [22]. Additionally, reverse osmosis systems are relatively expensive to install and maintain.

Calcium Hydroxide

High-level function: Maintain water quality
Function: Neutralize pH
Calcium hydroxide is not a viable option for neutralizing pH in a personal gray water plumbing system primarily because it can introduce excessive levels of calcium ions into the water [23], potentially causing scaling issues in pipes and fixtures over time. Furthermore, the effectiveness of calcium hydroxide in adjusting pH can be unpredictable.

9.3. Final solution space and alternate solutions

From the solution space, three different solutions are constructed. Below provided is the overall gist for each, the detailed differences can be referred to in the complete Morphological Table in Appendix A.

Solution 1

The focus of this solution is to reduce the overall footprint of the product. In that effort, means such as expandable bladder tanks are used. The system utilises a centralised collection tank. This ensures a much smaller footprint on the individual household level. The users are provided with water quality reports and tutorial videos for troubleshooting.

Solution 2

The focus of this solution was on maintainability and sustenance. Components like gravity drains and rain water integration in theory help make the system more sustainable. For user interaction a real time dashboard is proposed which gives readings from the various automated sensors and actuators chosen for this system.

Solution 3

This solution focuses on a completely distributed system. Instead of a central storage unit, each individual household unit will have its own discrete unit. This comes at the cost of a less intuitive user experience and space constraint restrictions on each component. This results in the selection of components with less space footprint over components with greater effect.

Finally, it was decided to go ahead with `Solution 1` as it provides space benefits over Solution 3 and cost benefits over Solution 2 while still full-filling the stipulated requirements. The centralised system removes the space constraint from other components allowing us to select the ones with the greatest efficacy. This solution also adapts to Barcelona, where the most common type of housing is small apartments with limited available space. It is also suitable for a country with limited purchasing power, and therefore, special emphasis is placed on the final cost of the product.

10. Risk Analysis

10.1. Risk Register

The risk register is part of the qualitative risk analysis, and maps identified risks to likelihoods and severity of impact. It also identifies people handling risks and taking mitigating actions [24]. The full risk register can be found in Appendix A.

10.2. Example Risk Analysis

A detailed risk analysis using the Bow-Tie method is shown in Figure 3 for the risk “Filter Clogging”. The risk describes the event that greywater filters inside the system congest during normal operation. In the Bow-Tie method different causes and consequences for a particular risk are visualized in order to “identify blockers to mitigate the risk” [25]. In the center, different events that lead to the risk are shown inside circles. On the left side, possible causes for the events are shown. The right side of the visualization contains identified effects or consequences.

The causes that lead to filter degradation have in common, that the filter material is worn down or damaged [26]. For example, abrasive particles, extreme temperatures and chemical reactions can affect the filter material negatively. Excessive flow rate or improper filter material can push particles through the filter media instead of trapping them, leading to clogging. The accumulation of particles can happen due to large or biologically grown up particles (e.g. algae or bacteria) contained in the greywater or improper filter installation and maintenance. Other causes can be

the chemical properties of the greywater like viscosity and chemical reactions like the formation of airbubbles [27] [28].

The effects range from medium to high impact. The reduction of water flow due to clogged filters decreases system efficiency and increases energy consumption of the pump to keep up the water pressure. It is also possible that contaminants build up, as a breeding ground for bacteria is developed in standing or slowly flowing water. This can lead to serious danger for users' health [29]. Increased tear on system components like pumps and pipes can lead to damage and failure of system components or the entire system and increased downtime and maintenance cost.

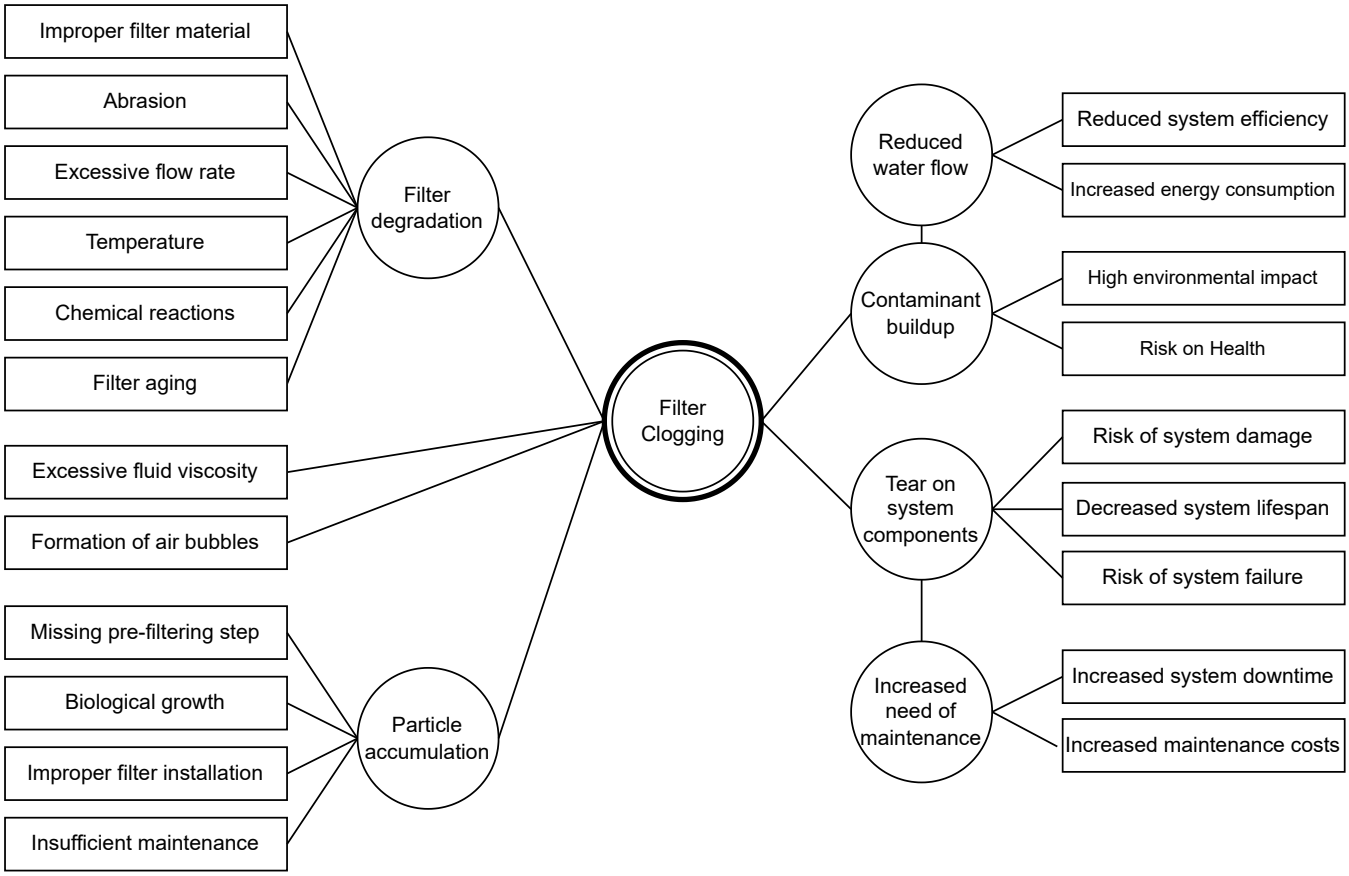

Figure 3: Bow-Tie Risk Analysis for Risk Filter Clogging

The ways to mitigate the effects of filter clogging can be directly derived from the Bow-Tie analysis. Regular cleaning and maintenance is necessary in order to maintain the filters in a good condition. Filters must be replaced either regularly on a scheduled interval or when the system detects clogged filters, e.g. through measured pressure differentials. This has to be done either by trained personnel or otherwise detailed instructions and training must be provided to the users [30] [31]. The filter material must be compatible with the expected chemical composition, temperature and flow rate of the greywater. A pre-filtering step should be included in order to remove large and abrasive particles and prevent biological growth.

Taking all the mitigation steps into account, the residual risk of filter clogging is very low. One possible scenario is that users could not follow the provided instructions and training for doing the filter replacement and filters would be installed improperly. Furthermore, there might be new secondary risks coming up, which include:

- A failure of the system that detects clogged filters,
- Trained personnel for filter replacement or system maintenance is unavailable,
- Chemical composition or temperature of greywater changes over time or due to unforeseen events,
- Failure in the pre-filtering step.

11. Test Plan

Testing helps to verify the operability, performance capability, and supportability of the greywater recycling system in either a simulated or real environment. A test plan is a set of selective tests that monitor the system in the above-mentioned criterion. There are different types of testing which are performed on a system:

1. Component Testing:

These testing are done on the individual components of the system, to monitor its performance and functionality based on its requirements as shown in Table 9.

Component	Filter Unit
Requirement	The system shall ensure that the water input fulfills the cleanliness requirements.
Description	**Title**: Gray Water Recycling Unit Filtration Quality **Goal**: The goal of this testing procedure is to confirm that the Gray Water Recycling Unit consistently and reliably maintains the specified cleanliness standards for the stored water. **Measurement**: The cleanliness of the water input will be evaluated against precise criteria, including measurements of turbidity levels, quantification of suspended solids, analysis for chemical contaminants, and detection of microorganisms. This evaluation will be performed using well-defined instruments and standardized testing methods. **Condition**: The Gray Water Recycling Unit will be installed and configured to match its typical operational setup. For instance, water temperature will be set at 25°C, flow rate at 10 liters per minute, and the system will be connected to a representative gray water source with a known contamination level of 200 ppm (parts per million) **Assumption**: It is assumed that the Gray Water Recycling Unit has been properly installed, calibrated, and maintained. The water source used for testing should be representative of typical gray water inputs. **Description**: A typical gray water sample is chosen and subjected to detailed cleanliness assessments, including turbidity measurement, suspended solids quantification, chemical contaminant analysis and microbiological testing for harmful microorganisms. These measurements are rigorously compared to predefined cleanliness requirements, based on regulatory standards. **Input**: Gray water sample (representative of typical input). **Output**: Test results for turbidity levels, suspended solids, chemical contaminants, and microorganisms.
Expected Response	The Gray Water Recycling Unit's cleanliness requirement is that the measured parameters of the incoming gray water, including turbidity, suspended solids, chemical contaminants, and microorganism levels, should meet or exceed the predefined cleanliness requirements.

Table 9: Component Test Table: Filter Unit

2. System Testing:

The entire system is a collection of many sub-systems and interfaces. These tests are done to verify whether it satisfies the overall requirements as shown in Table 10.

Component	Full System
Requirement	The system shall ensure that the water input fulfills the cleanliness requirements. The system shall maintain adequate water pressure. The system shall ensure that the gray water being used is kept odorless. The system shall inform the user when it requires maintenance. The system shall be able to store the recycled gray water. The system shall flush away the reused water after a threshold number of uses. The system must dump unusable water into the sewage.
Description	**Title**: Gray Water Recycling Unit Testing Procedure **Goal**: The goal of this test is to assess the Gray Water Recycling Unit's functionality and its compliance with specific requirements related to water quality, pressure maintenance, odor control, maintenance notification, water storage, flushing, and proper disposal of unusable water. **Measurement**: Input cleanliness, water pressure, odor control, maintenance notification, water storage, water flushing and unusable water disposal.

	Condition: The Gray Water Recycling Unit should be correctly installed in a small residential home, with all subunits (pump system, filter system, storage system, user interface) properly connected and powered. The system should be in a stable and safe operating condition. The water temperature will be set at 25°C, flow rate at 10 liters per minute, and the system will be connected to a representative gray water source with a known contamination level of 200 ppm (parts per million) **Assumption**: It is assumed that the Gray Water Recycling Unit is free from any pre-existing defects or malfunctions and that the testing environment accurately simulates typical residential usage conditions. Description: 1. Introducing water with known impurities to assess its ability to meet cleanliness requirements. 2. Monitoring water pressure to ensure it remains within the specified range during the test. 3. Assessing the system's capacity to keep gray water odorless during recycling. 4. Triggering various maintenance conditions and confirming that the user interface effectively communicates the need for maintenance. 5. Measuring the volume of stored recycled gray water to validate storage capacity. 6. Testing the system's ability to initiate water flushing when it reaches the predefined usage threshold. 7. Observing and confirming the correct routing of unusable water into the sewage system. **Input**: Contaminated water with known impurities, gray water for recycling, and water reaching the threshold for flushing. **Output**: Clean and recycled gray water meeting cleanliness requirements, maintained water pressure within the specified range, odorless gray water, accurate user notifications for maintenance, achieved water storage capacity, initiated water flushing at the predefined threshold, and proper disposal of unusable water into the sewage system.
Expected Response	the Gray Water Recycling Unit's cleanliness requirement is that the measured parameters of the incoming gray water, including turbidity, suspended solids, chemical contaminants, and microorganism levels, should meet or exceed the predefined cleanliness requirements. The ideal expected response to the Gray Water Recycling Unit test includes the system effectively reducing impurities in input water, maintaining consistent water pressure, ensuring odorless gray water, promptly notifying users of maintenance needs, achieving the specified water storage capacity, initiating water flushing at the designated threshold, and correctly disposing of unusable water into the sewage system, in full compliance with predefined requirements. This ideal response reflects the unit's efficient and reliable operation in small residential homes.

Table 10: System Test Table

12. Simulation Plan

Simulation aids in designing a graywater recycling system by allowing you to model and analyze its performance virtually. This helps optimize the design, predict outcomes, minimize costs, ensure regulatory compliance, and reduce risks before actual construction. The simulation plan of an important component of a graywater recycling system - the filter is laid out in Table 11. In a similar fashion the simulation plan for the complete system is shown in Table 12.

Section	Description
Simulation Goals	Optimize the performance of the filter unit in a gray water system. Specifically, maximize the efficiency of water filtration while minimizing operational and maintenance costs.
Components/Subsystems and their Interfaces	1. Gray Water Input: Gray water enters the system. 2. Filter Unit: Responsible for removing impurities and contaminants from gray water. 3. Pump: Facilitates the flow of gray water through the filter unit. 4. Outlet/Storage: Cleaned gray water or treated effluent is stored or distributed for reuse or discharge.
Model type	The model will be dynamic, continuous, deterministic, and possibly include some stochastic elements. It will simulate the behavior of the filter unit over time, considering the continuous flow of gray water and the variables affecting filtration efficiency
Inputs	1. Gray water quality parameters 2. Flow rate and volume of gray water entering the system. 3. Filter unit specifications (filter media type, surface area, maximum flow rate). 4. Pump specifications. 5. Cost data for components, operation, and maintenance.

Assumptions	1. The filter unit operates under steady-state conditions. 2. Gray water characteristics and flow rates are within expected ranges. 3. Filter performance is based on design specifications and laboratory tests. 4. Maintenance schedules and costs are predefined. 5. System variables are not subject to extreme fluctuations.
Initial conditions	At the start of the simulation, the system is assumed to be in a stable state. Gray water is flowing into the filter unit, and all system components are initially set to their specified operating conditions.
Expected outputs	1. Filtration efficiency, including the reduction in impurities, turbidity, and contaminants. 2. Water quality of the treated gray water. 3. Operational costs (e.g., energy consumption, maintenance, consumables). 4. System performance metrics (e.g., flow rates, pressure, filtration rate). 5. Comparison of different scenarios or configurations to optimize the filter unit's performance while minimizing costs. 6. Visualization of system behavior over time, including any alarms or events requiring attention.

Table 11: Simulation plan table for the component: Filters

Section	Description
Simulation Goals	Optimize the performance and efficiency of the graywater recycling system. This includes minimizing operational and maintenance costs while maximizing water recycling efficiency.
Components/Subsystems and their Interfaces	1. Pump Subsystem: This subsystem is responsible for pumping graywater from the source to the filter subsystem. 2. Filter Subsystem: Filters out impurities from graywater. 3. Storage Subsystem: Stores filtered graywater for reuse. 4. Monitoring Subsystem: Monitors the quality and quantity of water at various points in the system. 5. User Interface: Provides a user-friendly interface for system control and status monitoring.
Model type	The model used will be a dynamic, deterministic, continuous simulation model. It will account for changes over time, operate based on known parameters, and provide continuous data output.
Inputs	1. Flow rates and pressure data from the pump subsystem. 2. Filter specifications and performance data. 3. Storage tank capacity and dimensions. 4. Water quality data (initial graywater quality). 5. User preferences and usage patterns. 6. System operation and maintenance costs.
Assumptions	1. The unit operates under steady-state conditions. 2. Graywater quality remains relatively consistent. 3. Pump and filter efficiencies remain stable within the simulation timeframe. 4. User behavior follows typical patterns. 5. Maintenance and operational costs are constant.
Initial conditions	At t=0, We assume: 1. The storage tank is empty. 2. Pump subsystem is off. 3. The filter is clean. 4. Initial water quality is based on real-world data.
Expected outputs	1. Filtration efficiency, including the reduction in impurities, turbidity, and contaminants. 2. Water quality of the treated gray water. 3. Operational costs (e.g., energy consumption, maintenance, consumables). 4. System performance metrics (e.g., flow rates, pressure, filtration rate). 5. Total water recycled over a given time period. 6. User interface feedback and control options.

Table 12: Simulation plan table for the system

13. Final Design

The proposed design of a terrace-based greywater filtration system in a housing complex involves collecting greywater from apartment water outlets and recycling it for reuse. This sustainable solution minimizes water waste, reduces environmental impact, and provides residents with a reliable source of recycled water for non-potable purposes like irrigation, flushing, and more. The system includes filters and treatment processes to ensure water quality, and it is a cost-effective way to promote water conservation in a residential setting.

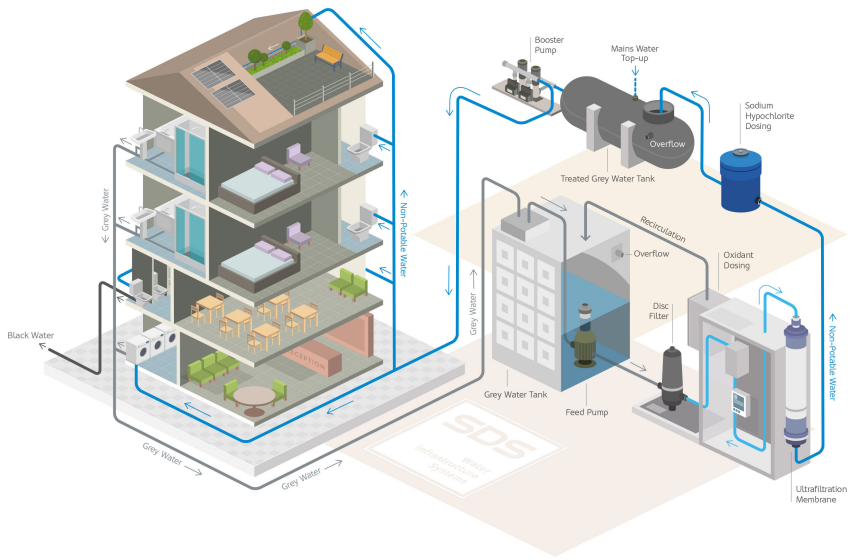

Figure 4: Final design schematic [32]

14. Conclusion

This report details the strategic approach to bolster Barcelona's drought resilience through water management solutions. The process began with concept generation based on the SDM, followed by requirement listing, conflict analysis, and functional decomposition. Utilizing the Morphological table, pivotal functions were determined. After a risk assessment using the bow-tie model, the system design was solidified. Test and simulation plans further validated the grey water filtration system's efficacy in addressing the city's challenges.

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A

Supporting Documents

Document Title	Document URL
Background & Introduction	https://tud365.sharepoint.com/:b:/s/ClimateChange-RC-7/ESDxs8nO_cBGsduxvPOChwEBI4l0iZ0mAv2J5AkACgTqeQ?e=RBa58h
Motivation & Need	https://tud365.sharepoint.com/:b:/s/ClimateChange-RC-7/EdSHZ_zC7IxOkv8pnx_BNnIB4AVhmWGSdOCrBXyYZfSiPw?e=fcOUlT
Objective Tree	https://tud365.sharepoint.com/:b:/s/ClimateChange-RC-7/ES3GI5tyKdBAgqzcW2yYkdkBM_EPB3uzLRd8wiKDvaKMCg?e=2jMQVs
System Definition Matrix	https://tud365.sharepoint.com/:b:/s/ClimateChange-RC-7/EbY7pXgU5O9DjLaUyfXzP-gBYLLexarMfNbmaz59y6118Q?e=2GrIeb
Trade-off Table	https://tud365.sharepoint.com/:b:/s/ClimateChange-RC-7/EViJ4ZGWtbJGm5wnjJckoFcBsQ_29mSbFbI6FFnnyx6lqw?e=iN6quE
Conflict Analysis	https://tud365.sharepoint.com/:b:/s/ClimateChange-RC-7/EcO9TuJnGIFNlrLbCxljD_gBkErd79EimjY59TF8yeuo9w?e=T4wnhy
Conflict Analysis Matrix	https://tud365.sharepoint.com/:b:/s/ClimateChange-RC-7/EfT1xnkvTZVEgs4wZ2646OsB-fDjXOkIa3rzwVCSGpmMuQ?e=a9F07l
List of Functions	https://tud365.sharepoint.com/:b:/s/ClimateChange-RC-7/ERWSazlnTWJPlhG9LKC7u_EB9OK1GUzM77GFb9t_iK4hPQ?e=7E4KI5
List of Requirements	https://tud365.sharepoint.com/:b:/s/ClimateChange-RC-7/Eb_5uZSwF0ZJru-ZLrO_kQwBXqIIDgHk_jiXMe8jnQH7Ng?e=McH45Q
List of Stakeholders	https://tud365.sharepoint.com/:b:/s/ClimateChange-RC-7/EYgPF9VEQPVDpcP78vbY8NoBPusCJsrCwnxSsOlaWgA-Zg?e=50id88
Functional Decomposition	https://tud365.sharepoint.com/:b:/s/ClimateChange-RC-7/ERQS7dqNNsZKmkD4Luz4_OQBOiPOsESnx4L3COB7QK43eA?e=BawBmS
Morphological Chart	https://tud365.sharepoint.com/:b:/s/ClimateChange-RC-7/EfoX_nBh26xPrUraeqRnTjoB3DLItUjlNGNodU_iB_GnFQ?e=2d4em9
Risk Register	https://tud365.sharepoint.com/:b:/s/ClimateChange-RC-7/EeDRtq9lTqJCumGvwMyoy88B9hvmaP7AThkKOqhG1jFnfQ?e=74WAMQ